BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate preferred embodiments of the invention.

[0011] FIG. 1 is a system architecture block diagram showing the self-sustaining neural-motor energy harvesting loop with a sensor 12, spiking neural network 14, motor 16, power management block 18, thermal harvester 20, piezoelectric element 22, ground reference 24, control logic block 26, energy store 28, and reflection feedback path 30 interconnected in a closed loop within an environment boundary 10.

[0012] FIG. 2 is a comparative energy balance chart showing control system energy depletion versus experimental system energy growth over 2,000 operational steps, plotted on a y-axis 32 and x-axis 34 with an origin point 38 at 50 mWh, a zero energy baseline 40, an experimental curve 36 with experimental curve label 42 and experimental legend box 46, and a control curve 44 with control curve label 50 and control legend box 48.

[0013] FIG. 3 is a schematic diagram of the spiking neural network 14 architecture showing the input layer 52, weight matrix 54, hidden layer 56 with leaky integrate-and-fire neurons and neuron detail section 58, output stage 60, spike output 62, membrane potential dynamics 64, spike vector output 66, and STDP learning rule box 68.

[0014] FIG. 4 is a circuit schematic of the energy harvesting subsystem within a circuit boundary 70 comprising a piezoelectric element 22 path with AC signal 72, inductor 74, rectifier 76, and smoothing capacitor 78, and a thermoelectric path with heat sink 80, TEG module 82, and boost converter 84, both feeding through an output regulator 86 and system output 88 to the spiking neural network 14.

[0015] FIG. 5 is a graph showing activity-dependent energy dynamics within a main graph area 90, including a sinusoidal harvest curve 92, quadratic cost curve 94, metabolic operating point 96, deficit zones 98, optimal operating range 100, surplus zone 102, optimal zone boundaries 104, governor feedback loop 106, energy-aware modulation sub-graph 132, operating point equations 134, lower crossover point 136, and upper crossover point 138.

[0016] FIG. 6 is a hardware reference design layout diagram showing the microcontroller 108, piezoelectric element 22, motor 16, thermistor 110, photoresistor 114, LED indicator 112, AC signal 72, control logic block 26, and energy store 28 with their interconnections.

[0017] FIG. 7 is a validation results summary comprising a claims validation table 116, an energy comparison table 118, an energy trajectory graph 120, a legend 122, and a summary box 124.

[0018] FIG. 8 is a reference architecture diagram of the energy-bounded recursive control system showing the signal flow from an environment input arrow 126 through sensor interface 12, cognitive modulation block 128, spiking neural network 14, motor 16, thermal harvester 20, analog-to-digital converter 130, piezoelectric element 22, and energy store 28, with reflection feedback 30, all within the environment boundary 10.